João Miguel Lourenço Ribeiro

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Current Positions

Aug 2024 - Instituto Superior Técnico – Universidade de Lisboa, Portugal
Professor Auxiliar (U.S. equivalent: Assistant Professor), Department of Mathematics

Aug 2024 - Instituto de Telecomunicações
Researcher

Education

2017-2021 **Imperial College London, UK**

Ph.D. in Computing
Thesis: Coding against synchronisation and related errors
Advisor: Mahdi Cheraghchi
Examiners: Cong Ling (Imperial College) and Michael Mitzenmacher (Harvard)

2015-2017 **ETH Zurich, Switzerland**

M.Sc. in Computer Science *(with distinction)*
Track: Theoretical Computer Science
Thesis: Challenges in information-theoretic secret-key agreement
(awarded the ETH Medal for outstanding M.Sc. theses)
Advisors: Ueli Maurer and Daniel Jost

2012-2015 **Instituto Superior Técnico – Universidade de Lisboa, Portugal**

B.Sc. in Applied Mathematics and Computation *(excellent)*

Previous Positions

Feb 2023 - July 2024 NOVA School of Science and Technology, Universidade Nova de Lisboa, Portugal
Professor Auxiliar (U.S. equivalent: Assistant Professor), Computer Science Department

Feb 2023 - July 2024 NOVA Laboratory for Computer Science and Informatics (NOVA LINCS)
Researcher

Aug 2021 - Jan 2023 Carnegie Mellon University, Pittsburgh, PA, USA
Post Doctoral Fellow, Computer Science Department

Hosted jointly by Vipul Goyal and Venkatesan Guruswami. Part of the Cryptography and Theory groups.

Visiting Positions

30 Jan 2024- 29 Feb 2024 Simons Institute for the Theory of Computing, University of California, Berkeley, CA, USA
Visiting Scientist

Invited long-term participant in the Error-Correcting Codes: Theory and Practice semester research program.

Feb 2020 - Mar 2020 University of Michigan, Ann Arbor, MI, USA
Visiting Scholar

Hosted by Mahdi Cheraghchi at the Computer Science and Engineering Department. Topics: information theory and theoretical computer science. Originally planned until May 2020, cut short due to the Covid-19 pandemic.

July 2019 - Aug 2019 University of Illinois at Urbana-Champaign, IL, USA
Visiting Scholar

Hosted by Olgica Milenkovic at the Coordinated Science Laboratory. Topic: coding theory for DNA-based data storage.

Feb 2019 - Apr 2019 Centre for Quantum Technologies, National University of Singapore, Singapore
Research Intern

Hosted by Divesh Aggarwal. Topics: pseudorandomness and information-theoretic cryptography.

July 2018 - Aug 2018 Centre for Quantum Technologies, National University of Singapore, Singapore
Research Intern

Hosted by Divesh Aggarwal. Topics: pseudorandomness and information-theoretic cryptography.

Participation in other invitation-based programs

April 2025 *Error-Correcting Codes: Theory and Practice Reunion*
Workshop at the Simons Institute for the Theory of Computing, University of California, Berkeley, CA, USA for invited long-term participants in the "Error-Correcting Codes: Theory and Practice" Spring 2024 program.

Dec 2024 *Dagstuhl Seminar 24511: Coding Theory and Algorithms for Emerging Technologies in Synthetic Biology*
Invited participant.

Selected Awards

2018 *ETH Medal*
Awarded by ETH Zurich for an outstanding M.Sc. thesis.

2015 *Excellence Scholarship & Opportunity Award*
Awarded by ETH Zurich to high potential M.Sc. students.

2015 *Professor Jaime Campos Ferreira Prize*
Awarded by the Department of Mathematics of Instituto Superior Técnico for outstanding performance in Mathematics.

2015 *Diploma of Academic Excellence*
Awarded by Instituto Superior Técnico.

2014-2015 *"New Talents in Mathematics" Scholarship*
Awarded by the Calouste Gulbenkian Foundations to 20 outstanding undergraduate students in mathematical subjects in Portugal.

2013-2014 *"New Talents in Mathematics" Scholarship*
Awarded by the Calouste Gulbenkian Foundations to 20 outstanding undergraduate students in mathematical subjects in Portugal.

Grants

1. ERC Starting Grant: LESYNCH – "Limits and efficiency of coding against synchronization errors" (EUR 1,497,000). DOI: 10.3030/101218842.

2. CMU-Portugal Exploratory Project: CODEFHE – "Fully-homomorphic encryption from post-quantum code-based assumptions" (EUR 67,000). In collaboration with Alexander Davidson (LASIGE, FCUL) and Aayush Jain (CMU). DOI: 10.54499/2024.12595.CMU. **Due to other commitments, I left the project on February 1, 2026, with Alexander Davidson taking on the role of PI.**

3. Protocol Labs Cryptonet Network Grant: "Stateless Distributed Randomness Generation" (USD 35,000). Co-PI with Chen-Da Liu Zhang (HSLU & Web3 Foundation), Elisaweta Masserova (CMU), Mark Simkin (Ethereum Foundation), Pratik Soni (U Utah), and Sri AravindaKrishnan Thyagarajan (U Sydney).

Patents

1. Olgica Milenkovic, Ryan Gabrys, João Ribeiro, Mahdi Cheraghchi. *Coded Trace Reconstruction*. United States Patent No. US 12,380,337 B2 (August 5, 2025).

Papers

All papers are available online at sites.google.com/site/joaorib94. DOIs or links to preprint versions are also listed below. Author ordering is almost always alphabetical (as usual in theoretical computer science). Works where author ordering has been chosen uniformly at random are signaled by ⓡ. Works with other non-alphabetical author ordering are signaled by ⓞ.

Journal papers

[J1] ⓞ João Pedro Leite, Guilherme Ramos, João Ribeiro, Sérgio Pequito, and David R. Matos. Byzantine-resilient random number generation via average consensus. *Systems & Control Letters*, 219:106649, 2027. 10.1016/j.sysconle.2026.106649.

[J2] Maciej Obremski, João Ribeiro, Lawrence Roy, François-Xavier Standaert, and Daniele Venturi. Improved reductions from noisy to bounded and probing leakages via hockey-stick divergences. *IEEE Transactions on Information Theory*, 2026. To appear. Extended version of [C13]. 10.1109/TIT.2026.3745906.

[J3] ⓞ Sahar Allahkaram, Francisco A. Monteiro, João Ribeiro, and Carlos Salema. Largely constrained noise-guessing decoding of short LDPC codes. *IEEE Open Journal of the Communications Society*, 2026. To appear. 10.1109/OJCOMS.2026.3725587.

[J4] Roni Con and João Ribeiro. Channels with input-correlated synchronization errors. *IEEE Transactions on Information Theory*, 72(8):5442–5472, 2026. Extended version of [C9]. 10.1109/TIT.2026.3694251.

[J5] Dean Doron, Jonathan Mosheiff, Nicolas Resch, and João Ribeiro. List-recovery of random linear codes over small fields. *IEEE Transactions on Information Theory*, 71(12):9548–9562, 2025. Extended version of [C7]. 10.1109/TIT.2025.3625861.

[J6] Dean Doron and João Ribeiro. Nearly-linear time seeded extractors with short seeds. *IEEE Transactions on Information Theory*, 71(11):9028–9054, 2025. 10.1109/TIT.2025.3605160.

[J7] Naresh Goud Boddu, Vipul Goyal, Rahul Jain, and João Ribeiro. Split-state non-malleable codes and secret sharing schemes for quantum messages. *IEEE Transactions on Information Theory*, 71(4):2838–2871, 2025. Extended version of [C11]. Preliminary version also presented as a contributed talk at QCRYPT 2023. 10.1109/TIT.2025.3530385.

[J8] Huck Bennett, Mahdi Cheraghchi, Venkatesan Guruswami, and João Ribeiro. Parameterized inapproximability of the minimum distance problem over all fields and the shortest vector problem in all ℓ_p norms. *SIAM Journal on Computing*, 53(5):1439–1475, 2024. Extended version of [C16]. 10.1137/23M1573021.

[J9] ⓪ Ananthan Nambiar, Chao Pan, Vishal Rana, Mahdi Cheraghchi, João Ribeiro, Sergei Maslov, and Olgica Milenkovic. Semi-quantitative group testing for efficient and accurate qPCR screening of pathogens with a wide range of loads. *BMC bioinformatics*, 25(1):195, 2024. 10.1186/s12859-024-05798-3.

[J10] Mahdi Cheraghchi and João Ribeiro. Simple codes and sparse recovery with fast decoding. *SIAM Journal on Discrete Mathematics*, 37(2):612–631, 2023. Extended version of [C29]. 10.1137/21M1465354.

[J11] Ryan Gabrys, Venkatesan Guruswami, João Ribeiro, and Ke Wu. Beyond single-deletion correcting codes: Substitutions and transpositions. *IEEE Transactions on Information Theory*, 69(1):169–186, 2023. Extended version of [C19]. 10.1109/TIT.2022.3202856.

[J12] Gianluca Brian, Antonio Faonio, João Ribeiro, and Daniele Venturi. Short non-malleable codes from related-key secure block ciphers, revisited. *IACR Transactions on Symmetric Cryptology*, 2022(3):1–19, Sep. 2022. This work has also been presented at the 2023 Fast Software Encryption (FSE) Workshop. 10.46586/tosc.v2022.i3.1-19.

[J13] Gianluca Brian, Antonio Faonio, Maciej Obremski, João Ribeiro, Mark Simkin, Maciej Skórski, and Daniele Venturi. The mother of all leakages: How to simulate noisy leakages via bounded leakage (almost) for free. *IEEE Transactions on Information Theory*, 68(12):8197–8227, 2022. Extended version of [C23]. 10.1109/TIT.2022.3193848.

[J14] Divesh Aggarwal, Maciej Obremski, João Ribeiro, Mark Simkin, and Luisa Siniscalchi. Privacy amplification with tamperable memory via non-malleable two-source extractors. *IEEE Transactions on Information Theory*, 68(8):5475–5495, 2022. 10.1109/TIT.2022.3167404.

[J15] Mahdi Cheraghchi, Joseph Downs, João Ribeiro, and Alexandra Veliche. Mean-based trace reconstruction over oblivious synchronization channels. *IEEE Transactions on Information Theory*, 68(7):4272–4281, 2022. Extended version of [C22]. 10.1109/TIT.2022.3157383.

[J16] Mahdi Cheraghchi and João Ribeiro. Non-asymptotic capacity upper bounds for the discrete-time Poisson channel with positive dark current. *IEEE Communications Letters*, 25(12):3829–3832, 2021. 10.1109/LCOMM.2021.3120706.

[J17] Mahdi Cheraghchi and João Ribeiro. An overview of capacity results for synchronization channels. *IEEE Transactions on Information Theory*, 67(6):3207–3232, 2021. 10.1109/TIT.2020.2997329.

[J18] Mahdi Cheraghchi, Ryan Gabrys, Olgica Milenkovic, and João Ribeiro. Coded trace reconstruction. *IEEE Transactions on Information Theory*, 66(10):6084–6103, 2020. Extended version of [C28]. 10.1109/TIT.2020.2996377.

[J19] Mahdi Cheraghchi and João Ribeiro. Sharp analytical capacity upper bounds for sticky and related channels. *IEEE Transactions on Information Theory*, 65(11):6950–6974, Nov 2019. Extended version of [C30]. 10.1109/TIT.2019.2920375.

[J20] Mahdi Cheraghchi and João Ribeiro. Improved upper bounds and structural results on the capacity of the discrete-time Poisson channel. *IEEE Transactions on Information Theory*, 65(7):4052–4068, July 2019. Extended version of [C31]. 10.1109/TIT.2019.2896931.

[J21] ⓪ João Ribeiro, André Souto, and Paulo Mateus. Quantum blind signature with an offline repository. *International Journal of Quantum Information*, 13(02):1550016, 2015. Undergraduate research. 10.1142/S0219749915500161.

Conference papers

[C1] Roni Con, Dean Doron, and João Ribeiro. The insertion list-decoding capacity and an improved bound on the deletion list-decoding capacity. To appear at ITCS 2027. https://arxiv.org/abs/2607.03989.

[C2] Julien Béguinot, Ananta Mukherjee, Maciej Obresmki, João Ribeiro, Lawrence Roy, François-Xavier Standaert, and Daniele Venturi. Simulating noisy leakage with bounded leakage: Simpler, better, faster. To appear at Asiacrypt 2026. https://eprint.iacr.org/2026/357.

[C3] ⓪ Cécile Bouette, Samuel Pearson, Roni Con, and João Ribeiro. Capacity of additive-noise sticky channels. In *2026 IEEE Information Theory Workshop (ITW)*, 2026. To appear. https://arxiv.org/abs/2608.01433.

[C4] ⓪ Yuan-Pon Chen, Olgica Milenkovic, João Ribeiro, and Jin Sima. Correcting contextual deletions in DNA nanopore readouts. In *2026 IEEE Information Theory Workshop (ITW)*, 2026. To appear. https://arxiv.org/abs/2602.05072.

[C5] ⓪ Martim Pinto and João Ribeiro. Improved capacity upper bounds for the deletion channel using a parallelized Blahut-Arimoto algorithm. In *2026 IEEE International Symposium on Information Theory (ISIT)*, 2026. To appear. https://arxiv.org/abs/2604.05867.

[C6] Chen-Da Liu-Zhang, Elisaweta Masserova, João Ribeiro, and Sri AravindaKrishnan Thyagarajan. Game theory does not always help: The case of statistical coin tossing. In *Advances in Cryptology – EUROCRYPT 2026*, pages 488–514, 2026. 10.1007/978-3-032-25324-8_17.

[C7] Dean Doron, Jonathan Mosheiff, Nicolas Resch, and João Ribeiro. List-recovery of random linear codes over small fields. In *RANDOM 2025*, pages 57:1–57:18, 2025. 10.4230/LIPIcs.APPROX/RANDOM.2025.57.

[C8] Omar Alrabiah, Jesse Goodman, Jonathan Mosheiff, and João Ribeiro. Low-degree polynomials are good extractors. In *RANDOM 2025*, pages 38:1–38:25, 2025. 10.4230/LIPIcs.APPROX/RANDOM.2025.38.

[C9] Roni Con and João Ribeiro. Channels with input-correlated synchronization errors. In *2025 IEEE International Symposium on Information Theory (ISIT)*, 2025. 10.1109/ISIT63088.2025.11195433.

[C10] Chen-Da Liu-Zhang, Elisaweta Masserova, João Ribeiro, Pratik Soni, and Sri AravindaKrishnan Thyagarajan. Efficient distributed randomness generation from minimal assumptions where PArties Speak Sequentially Once. In *Advances in Cryptology – EUROCRYPT 2025*, pages 176–206, 2025. Also presented as a contributed talk at TPMPC 2025. 10.1007/978-3-031-91092-0_7.

[C11] Naresh Goud Boddu, Vipul Goyal, Rahul Jain, and João Ribeiro. Split-state non-malleable codes and secret sharing schemes for quantum messages. In *2024 Theory of Cryptography Conference (TCC 2024)*, pages 60–93, 2024. 10.1007/978-3-031-78017-2_3. Preliminary version presented as a contributed talk at QCRYPT 2023.

[C12] Alper Çakan, Vipul Goyal, Chen-Da Liu Zhang, and João Ribeiro. Unbounded leakage-resilience and intrusion-detection in a quantum world. In *2024 Theory of Cryptography Conference (TCC 2024)*, pages 159–191, 2024. 10.1007/978-3-031-78017-2_6. Also presented as a contributed talk at TQC 2024.

[C13] Maciej Obremski, João Ribeiro, Lawrence Roy, François-Xavier Standaert, and Daniele Venturi. Improved reductions from noisy to bounded and probing leakages via hockey-stick divergences. In *Advances in Cryptology – CRYPTO 2024*, pages 461–491, 2024. 10.1007/978-3-031-68391-6_14.

[C14] Chen-Da Liu-Zhang, Elisaweta Masserova, João Ribeiro, Pratik Soni, and Sri AravindaKrishnan Thyagarajan. Improved YOSO randomness generation with worst-case corruptions. In *2024 International Conference on Financial Cryptography and Data Security (FC 2024)*, pages 73–89, 2025. 10.1007/978-3-031-78679-2_4.

[C15] Alper Çakan, Vipul Goyal, Chen-Da Liu Zhang, and João Ribeiro. Computational quantum secret sharing. In *18th Conference on the Theory of Quantum Computation, Communication and Cryptography (TQC 2023)*, pages 4:1–4:26, 2023. 10.4230/LIPIcs.TQC.2023.4.

[C16] Huck Bennett, Mahdi Cheraghchi, Venkatesan Guruswami, and João Ribeiro. Parameterized inapproximability of the minimum distance problem over all fields and the shortest vector problem in all ℓ_p norms. In *55th Annual ACM Symposium on Theory of Computing (STOC 2023)*, pages 553–566, 2023. 10.1145/3564246.3585214.

[C17] Vipul Goyal, Chen-Da Liu Zhang, Justin Raizes, and João Ribeiro. Asynchronous multi-party quantum computation. In *14th Innovations in Theoretical Computer Science Conference (ITCS 2023)*, pages 62:1–62:22, 2023. 10.4230/LIPIcs.ITCS.2023.62.

[C18] Divesh Aggarwal, Eldon Chung, Maciej Obremski, and João Ribeiro. On secret sharing, randomness, and random-less reductions for secret sharing. In *Theory of Cryptography Conference (TCC) 2022*, pages 327–354, 2022. 10.1007/978-3-031-22318-1_12.

[C19] Ryan Gabrys, Venkatesan Guruswami, João Ribeiro, and Ke Wu. Beyond single-deletion correcting codes: Substitutions and transpositions. In *RANDOM 2022*, pages 8:1–8:17, 2022. 10.4230/LIPIcs.APPROX/RANDOM.2022.8.

[C20] ⓡ Jesper Buus Nielsen, João Ribeiro, and Maciej Obremski. Public randomness extraction with ephemeral roles and worst-case corruptions. In *Advances in Cryptology – CRYPTO 2022*, pages 127–147, 2022. 10.1007/978-3-031-15802-5_5.

[C21] Omar Alrabiah, Eshan Chattopadhyay, Jesse Goodman, Xin Li, and João Ribeiro. Low-degree polynomials extract from local sources. In *49th International Colloquium on Automata, Languages, and Programming (ICALP 2022)*, pages 10:1–10:20, 2022. 10.4230/LIPIcs.ICALP.2022.10.

[C22] Mahdi Cheraghchi, Joseph Downs, João Ribeiro, and Alexandra Veliche. Mean-based trace reconstruction over practically any replication-insertion channel. In *2021 IEEE International Symposium on Information Theory (ISIT)*, pages 2459–2464, 2021. 10.1109/ISIT45174.2021.9518161.

[C23] Gianluca Brian, Antonio Faonio, Maciej Obremski, João Ribeiro, Mark Simkin, Maciej Skórski, and Daniele Venturi. The mother of all leakages: How to simulate noisy leakages via bounded leakage (almost) for free. In *Advances in Cryptology – Eurocrypt 2021*, pages 408–437, 2021. 10.1007/978-3-030-77886-6_14.

[C24] Divesh Aggarwal, Siyao Guo, Maciej Obremski, João Ribeiro, and Noah Stephens-Davidowitz. Extractor lower bounds, revisited. In *RANDOM 2020*, pages 1:1–1:20, 2020. 10.4230/LIPIcs.APPROX/RANDOM.2020.1.

[C25] Abhishek Agarwal, Olgica Milenkovic, Srilakshmi Pattabiraman, and João Ribeiro. Group testing with runlength constraints for topological molecular storage. In *2020 IEEE International Symposium on Information Theory (ISIT)*, pages 132–137, 2020. 10.1109/ISIT44484.2020.9174502.

[C26] Divesh Aggarwal, Maciej Obremski, João Ribeiro, Luisa Siniscalchi, and Ivan Visconti. How to extract useful randomness from unreliable sources. In *Advances in Cryptology – Eurocrypt 2020*, pages 343–372, 2020. 10.1007/978-3-030-45721-1_13.

[C27] Divesh Aggarwal, Ivan Damgård, Jesper Buus Nielsen, Maciej Obremski, Erick Purwanto, João Ribeiro, and Mark Simkin. Stronger leakage-resilient and non-malleable secret sharing schemes for general access structures. In *Advances in Cryptology – CRYPTO 2019*, pages 510–539, 2019. 10.1007/978-3-030-26951-7_18.

[C28] ⓞ Mahdi Cheraghchi, João Ribeiro, Ryan Gabrys, and Olgica Milenkovic. Coded trace reconstruction. In *2019 IEEE Information Theory Workshop (ITW)*, pages 1–5, 2019. 10.1109/ITW44776.2019.8989261.

[C29] Mahdi Cheraghchi and João Ribeiro. Simple codes and sparse recovery with fast decoding. In *2019 IEEE International Symposium on Information Theory (ISIT)*, pages 156–160, 2019. 10.1109/ISIT.2019.8849702.

[C30] Mahdi Cheraghchi and João Ribeiro. Sharp analytical capacity upper bounds for sticky and related channels. In *2018 56th Annual Allerton Conference on Communication, Control, and Computing (Allerton)*, pages 1104–1111, 2018. 10.1109/ALLERTON.2018.8636009.

[C31] Mahdi Cheraghchi and João Ribeiro. Improved capacity upper bounds for the discrete-time Poisson channel. In *2018 IEEE International Symposium on Information Theory (ISIT)*, pages 1769–1773, 2018. 10.1109/ISIT.2018.8437514.

[C32] Daniel Jost, Ueli Maurer, and João L. Ribeiro. Information-theoretic secret-key agreement: The asymptotically tight relation between the secret-key rate and the channel quality ratio. In *2018 Theory of Cryptography Conference (TCC)*, pages 345–369, 2018. 10.1007/978-3-030-03807-6_13.

[C33] Ueli Maurer and João Ribeiro. New perspectives on weak oblivious transfer. In *2016 IEEE International Symposium on Information Theory (ISIT)*, pages 790–794, 2016. 10.1109/ISIT.2016.7541407.

Preprints

[M1] Dean Doron, Tal Leonov, Jonathan Mosheiff, Henrique Navas, Nicolas Resch, and João Ribeiro. Discrepancy for random linear codes. https://arxiv.org/abs/2606.24471.

[M2] Ⓞ Keshav Goyal, Samuel Pearson, João Ribeiro, and Serge Kas Hanna. The synthesis-sequencing channel for DNA-based data storage. https://arxiv.org/abs/2606.07216.

[M3] Alex Davidson, Nuno Nogueira, Samuel Pearson, and João Ribeiro. Streamlined symmetric private information retrieval via Rényi divergence. https://eprint.iacr.org/2026/1051.

[M4] Divesh Aggarwal, Zihan Li, Saswata Mukherjee, Maciej Obremski, and João Ribeiro. Complete characterization of randomness extraction from DAG-correlated sources. https://eprint.iacr.org/2026/066.

Selected Talks

1. *Packing oranges at random.*

   Invited colloquium talk at the Department of Mathematical Sciences, Faculty of Sciences, University of Lisbon, Portugal, September 2026.

2. *Sampling codes with memory and the insertion list-decoding capacity.*

   Invited talk at the ICE Seminar, Technical University of Munich, Germany, June 2026.

3. *Coding against correlated synchronization errors.*

   Invited talk at the International Zurich Seminar on Information and Communication, Zurich, Switzerland, February 2026.

4. *Leakage-resilient and non-malleable secret sharing.*

   Invited keynote talk at the Project Hermes' workshop on secret sharing schemes, Tarragona, Spain, May 2025.

5. *Low-degree polynomials are good extractors.*

   - Invited talk at the AlgoTheory seminar, National University of Singapore. August 2025.
   - Error-Correcting Codes: Theory and Practice Reunion Workshop at the Simons Institute for the Theory of Computing, UC Berkeley. April 2025.

6. *Randomness as a computational resource.*

   Invited talk at the World Logic Day workshop of the Portuguese Logic Society at Instituto Superior Técnico, Universidade de Lisboa, Portugal, January 2025.

7. *"Noisy" versus "Bounded" leakage.*

   - Invited talk at Portugal Crypto Day, December 2024.
   - Invited talk at Cryptography Seminar, Carnegie Mellon University, November 2024. Recording at https://youtu.be/_3IB5qn0ThQ?si=JKofwkVDCs1n3ZZW.

8. *Parameterized hardness of coding and lattice problems.*

- Invited talk at Theory of Computing Seminar, Faculdade de Ciências, University of Lisbon, November 2023.
- Invited talk at Talks@DCC Seminar, Faculdade de Ciências, University of Porto, May 2023.

9. *Public randomness extraction with ephemeral roles and worst-case corruptions.*
   - Invited talk at Cryptography Seminar, ETH Zurich, July 2023.
   - CRYPTO 2022, August 2022.
     Recording at https://www.youtube.com/watch?v=TGRUGoeRA1g
   - Invited talk at Indian Institute of Science – Microsoft Research Lecture Series, August 2022.
     Recording at https://www.youtube.com/watch?v=zob_q-ck8Qo
10. *Low-degree polynomials, local sources, and a curious log factor.*
    CMU Theory Lunch, March 2022.
    Recording at https://www.youtube.com/watch?v=eviaYIt_S6M
11. *The mother of all leakages: How to simulate noisy leakages via bounded leakage (almost) for free.*
    - Invited talk at Logic and Computation Seminar, Instituto Superior Técnico, University of Lisbon, June 2022.
    - Special in-person workshop at the 2021 Theory of Cryptography Conference, November 2021.
12. *Extractor lower bounds, revisited.*
    Random 2020, August 2020.
    Recording at https://www.youtube.com/watch?v=JpHcqsqMFr0
13. *How to extract useful randomness from unreliable sources.*
    Eurocrypt 2020, May 2020.
    Recording at https://www.youtube.com/watch?v=15zsUxU9y2o
14. *Coded and uncoded trace reconstruction.*
    Invited talk at the Shannon Channel (hosted by Salim El Rouayheb), September 2019.
    Recording at https://www.youtube.com/watch?v=mMEeGD6a0qI
15. *Information-theoretic secret-key agreement and classical bound entanglement.*
    Invited talk at Quantum Computation and Information Seminar, Instituto Superior Técnico, University of Lisbon, February 2019.

Teaching Experience

As Main/Co-lecturer

Spring 2026 Main lecturer for the Theory of Computation course at Instituto Superior Técnico, Universidade de Lisboa. This course is taught to 2nd year BSc students in Computer Science and Engineering.

Fall 2025 Main lecturer for the Introduction to Coding Theory course at Instituto Superior Técnico, Universidade de Lisboa. This course is taught to MSc students in Applied Mathematics and Computation.
Median student evaluation score: 9/9 (Excellent)

Spring 2024 Main lecturer for the Cryptography and Security Protocols course at Instituto Superior Técnico, Universidade de Lisboa. This course is taught to MSc students in Applied Mathematics and Computation, Computer Science, and others.
Median student evaluation score: 9/9 (Excellent)
Awarded a Teaching Excellence Diploma by IST-ULisboa for this course.

Fall 2024 Main lecturer for the Algorithms and Computational Modeling course at Instituto Superior Técnico, Universidade de Lisboa. This course is taught to 2nd year BSc students in Biomedical Engineering and Biological Engineering, and to some 3rd year BSc students in Applied Mathematics and Computation.

Fall 2024 Co-lecturer for the Elements of Programming course at Instituto Superior Técnico, Universidade de Lisboa. This course is taught to 1st year Applied Mathematics and Computation BSc students.

Spring 2024 Main lecturer (regente) for the Theory of Computation course at the NOVA School of Science and Technology, Universidade Nova de Lisboa (FCT-UNL). This course is taught to 2nd year Computer Science BSc students.
Summary of evaluation by students: *Completely Satisfied (6/6)* (37%), *Very Satisfied (5/6)* (30%).

Fall 2023 Co-lecturer for the Introduction to Programming course at the NOVA School of Science and Technology, Universidade Nova de Lisboa (FCT-UNL). This course is taught to 1st year Computer Science BSc students.
Summary of evaluation by students: *Completely Satisfied (6/6)* (17%), *Very Satisfied (5/6)* (42%).

Spring 2023 Lecturer for the "Codes and Lattices in Cryptography" informal 5-lecture mini-course at the NOVA School of Science and Technology, Universidade Nova de Lisboa (FCT-UNL). This mini-course was aimed mostly at Masters and PhD students and also faculty of the Department of Mathematics of FCT-UNL. The full set of lecture notes can be found at the course webpage.

Spring 2023 Co-lecturer for the Theory of Computation course at the NOVA School of Science and Technology, Universidade Nova de Lisboa (FCT-UNL). This course is taught to 2nd year Computer Science BSc students.

As Teaching Assistant

Fall 2020 Tutor for the Mathematics I course at Imperial College London. Duties include leading a weekly small-group tutorial and grading weekly assessed coursework.

Fall 2018 Graduate Teaching Assistant for the Information & Coding Theory and Algorithms II courses at Imperial College London. Duties included teaching weekly exercise classes, grading midterms, and designing coursework.

Fall 2017 Graduate Teaching Assistant for the Information & Coding Theory course at Imperial College London. Duties included teaching weekly exercise classes and grading midterms.

Fall 2016 Teaching Assistant for the Discrete Mathematics course at ETH Zurich. Duties included leading a weekly small-group tutorial and grading weekly homework.

Fall 2015 Teaching Assistant for the Discrete Mathematics course at ETH Zurich. Duties included leading a weekly small-group tutorial and grading weekly homework.

Mentoring and Supervision

Post doctoral researchers

1. Tovohery Randrianarisoa, May 2026 – present.
2. Cécile Bouette, March 2026 – present.

PhD students

1. Abhinav Vaishya, May 2026 – present.
2. Samuel Pearson, Feb 2026 – present.
3. Henrique Navas, Sept 2025 – present. Co-advised with Miguel Couceiro and Victor Lagerkvist.

Gulbenkian's New Talents in Mathematics Program

This section lists students mentored through the Calouste Gulbenkian Foundation's New Talents in Mathematics (*Novos Talentos em Matemática*) program. This program awards fellowships to outstanding students studying mathematical subjects at Portuguese universities. The fellowship includes a year-long research project.

1. Laura Afonso Guerreiro (Math, IST-UL), Nov 2024 – Sept 2025. Topic: Game-theoretically fair coin tossing protocols.
2. Mariana Rio Costa (Math, IST-UL), Nov 2023 – Sept 2024. Topic: Complexity of computational problems on point lattices.

MSc theses

1. Rita Pessoa (CS, IST-UL). Sept 2025 – Oct 2026. Co-advised with Ricardo Chaves.
2. Francisco Capelo (Math, IST-UL), Feb 2025 – Oct 2025.
3. Miguel Graça (Math, IST-UL), Feb 2025 – Oct 2025.
4. Pedro Gomes (CS, IST-UL), Sept 2024 – Oct 2025. Co-advised with Ricardo Chaves.
5. Samuel Pearson (Math, IST-UL), Sept 2024 – Oct 2025. Co-advised with Alexander Davidson.
6. Francisco Costa (CS, FCT-UNL), Feb 2024 – April 2025. Co-advised with Alexander Davidson.
7. Diogo Ramos (CS, FCT-UNL), Feb 2023 – Sept 2024. Co-advised with Alexander Davidson.
8. Gonçalo Cavaco (CS, FCT-UNL), Feb 2023 – April 2024.

MSc research projects

MSc research project students that have followed up with an MSc thesis under my supervision are listed in the previous section.

1. Tiago Mourão (Math, IST-UL). Feb 2026 – July 2026.
2. Luís Correia (Math, IST-UL). Feb 2026 – July 2026.

BSc research and final projects

1. Martim Pinto (Math BSc final project, IST-UL), 2026.
2. Martim Pinto (independent research project, Math BSc, IST-UL), 2024–2026.
   Next steps: Photonics Engineering MSc at Ghent University.
3. Daniel Bartolomeu (Math BSc final project, IST-UL), 2025.
4. Francisco Relvas (Math BSc final project, IST-UL), 2025.
5. Mariana Rio Costa (Math BSc final project, IST-UL), 2025.
   Next steps: Logic MSc at the University of Amsterdam.
6. Pedro Bezerra da Costa (Math BSc final project, IST-UL), 2024.
   Next steps: MFoCS MSc at the University of Oxford.
7. Rui Zhu Wang (Math BSc final project, IST-UL), 2024.
   Next steps: MFoCS MSc at the University of Oxford.
8. Tomás Oliveira (Math BSc final project, IST-UL), 2024.
9. Diogo Carvalho (CS BSc research project, FCT-UNL), 2024. Co-advised with Alexander Davidson.
10. Arda Aydın (BSc student, Boğaziçi University), remote research project, 2021. Co-advised with Venkatesan Guruswami.
    Next steps: PhD student in the ECE Department of the University of Maryland, College Park, MD, USA.

MSc/PhD committee membership

PhD committee membership

1. External member of PhD committee for Amineh Sakhaie (Faculdade de Ciências – Universidade de Lisboa).
2. External member of PhD committee for Francesca Ugazio (Faculdade de Ciências – Universidade de Lisboa).
3. External member of PhD committee for Alexandra Veliche (University of Michigan – Ann Arbor). Defended May 27, 2025.
4. Member of PhD defense committee for Manuel Santos (IST – Universidade de Lisboa). Defended January 16, 2025.

MSc committee membership

1. Member of MSc thesis defense committee for Lourenço Abecasis (Instituto Superior Técnico, Universidade de Lisboa). Thesis title: UC Bit Commitment with Communicating Malicious PUFs. Supervised by Paulo Mateus and Chrysoula Vlachou. Defended December 10, 2024.
2. Member of MSc thesis defense committee for Bernardo Ramalho (Faculdade de Engenharia, University of Porto). Thesis title: Secure Machine Learning via Homomorphic Encryption. Supervised by Manuel Barbosa and Alberto Pedrouzo. Defended July 17, 2023.
3. Member of MSc thesis defense committee for Henrique Navas (Instituto Superior Técnico, Universidade de Lisboa). Thesis title: Weakly Non-Computable Processes Described by Evolving Recursive Functions. Supervised by José Félix Costa. Defended July 6, 2023.

Academic Service

- Co-organizer of the CMU Cryptography Seminar (08/2021 – 01/2023).
- **Member of the Conference Program Committee for:**
  1. The 2027 IEEE Information Theory Workshop (ITW 2027).
  2. Eurocrypt 2027.
  3. Asiacrypt 2026.
  4. The 23rd IACR Theory of Cryptography Conference (TCC 2025).
  5. The 45th Annual International Cryptology Conference (CRYPTO 2025).
  6. The 28th International Conference on Randomization and Computation (RANDOM 2024).
  7. The 5th Conference on Information-Theoretic Cryptography (ITC 2024).
  8. The 21st IACR Theory of Cryptography Conference (TCC 2023).
  9. The 4th Conference on Information-Theoretic Cryptography (ITC 2023).
- **External reviewer for the following conferences:** CCC (2026), CiE (Computability in Europe, 2023), CRYPTO (2020, 2021), Eurocrypt (2019, 2020, 2021, 2022), FOCS (2019, 2020, 2023, 2024), ICALP (2022), ISIT (2017–2026), Conference on Information-Theoretic Cryptography (ITC, 2020), ITCS (2019, 2021, 2022, 2024, 2025), ITW (2019, 2020, 2021, 2022), SODA (2020, 2021), STOC (2018, 2021, 2022, 2025), TCC (2018, 2019, 2021), Conference on Security and Cryptography for Networks (SCN, 2020).
- **Reviewer for the following journals:** IEEE Transactions on Information Theory, IEEE Transactions on Communications, IEEE Transactions on Molecular, Biological, and Multi-Scale Communications, Discrete Applied Mathematics.
- Reviewer for AMS Mathematical Reviews (2022 – present).
- Reviewer of funding proposals for the Israel Science Foundation (ISF).

Dissemination and Outreach

Presentations to academic audiences

1. Invited 3 hour mini-course on coding theory at the 2026 Lisbon Young Mathematicians' Conference, aimed at PhD students in Mathematics (NOVA FCT, September 2026).

Presentations to general audiences

1. Invited presentations on error-correcting codes to middle-school students within programs organized by TreeTree2 (Advanced School of Mathematics, Summer School), a non-profit that aims to democratize access to STEM education in Portugal (March and July 2026).

2. Invited presentation on mathematics and its applications to communication and data processing systems in the "Mathematics and Hope: Levers of Innovation" event organized by the Lisbon Academy of Sciences, Centro Internacional de Matemática (CIM), and Center of Mathematics of the University of Porto (CMUP) to commemorate Pi Day (March 2026). This event included a round table discussion on the future of basic research in Portugal with all speakers and Fernando Alexandre, the Minister of Education, Science, and Innovation of Portugal.

   The recordings of the talk can be found here, and the recording of the round-table discussion can be found here.

3. Invited presentation on secret sharing in the "Time2Talk" workshop organized by the Mathematics Students Association (NMATH) of Técnico-ULisboa (March 2026).

4. Invited presentation on error-correcting codes in the "New Talents in Mathematics" workshop at the Calouste Gulbenkian Foundation, aimed at BSc and MSc students with fellowships from the "New Talents in Mathematics" program (December 2025).

5. Invited presentation on cryptography in the "Mathematics Week" of Colégio Valsassina, aimed at high school students (November 2024).

6. Invited presentation on cryptography in MatNova 2024, a Summer school about mathematics for high school students hosted at the NOVA School of Science and Technology (September 2024).

Committee memberships

1. Scientific committee member for ToPAS 2024, a programming competition for high school students in Portugal.

Writings focused on scientific dissemination and outreach

1. *Mathematics and the age of information*, an invited contribution in a monograph of the Lisbon Academy of Sciences (available here) following the "Mathematics and Hope: Levers of Innovation" Pi Day event in Porto (March 2026) where I gave an invited talk (focusing on a different subject). This monograph includes an afterword by Fernando Alexandre, the current Minister of Education, Science, and Innovation.

   The main goal of this monograph and of the associated event was to showcase to a wider public how basic research in mathematics-adjacent fields has, time and time again, played a surprising pivotal role in major practical breakthroughs decades or hundreds of years afterwards. These dissemination efforts arose as a reaction to the decision by the Portuguese Government to merge the institutions that fund scientific research (FCT) and innovation (ANI) into a single institution (AI^2), with virtually no consultation by scientists and without presenting a concrete plan for the future of scientific funding.

2. Como poupar testes de rastreio: A testagem em grupos como introdução ao método probabilístico (in Portuguese). Appears in the Gazeta de Matemática, nr. 203, 2024, a publication of the Portuguese Mathematical Society.

3. COVID-19 group testing annotated bibliography, edited in collaboration with Laura Balzano, Kyle Gilman, Matthew Malloy, Ivo Stoepker, and Yutong Wang, 2020.